

Nathan Bakare

+1 (832) 946-6005 | Nbakare@cougarnet.uh.edu | [linkedin.com/in/nathan-bakare](https://www.linkedin.com/in/nathan-bakare) | github.com/Thenathanb

EDUCATION

University of Houston

Houston, TX

Bachelor of Science in Computer Engineering | GPA: 3.87/4.0

Expected May 2028

Relevant Coursework: Circuit Analysis, Data Structures & Algorithms, Fundamentals of Software Engineering, Operating Systems, Database Systems, Object Oriented Programming

TECHNICAL SKILLS

Languages: Python, Java, C, HTML/CSS, JavaScript, TypeScript, SQL

Frameworks/Libraries: React, Next.js, Node.js, Express.js, FastAPI, Flask, PyTorch, XGBoost, Tailwind CSS

Tools/Platforms: Git, GitHub, Docker, MLflow, Airflow, GraphQL API, Langflow, OpenSearch, Docling, pytest, Vitest

EXPERIENCE

IBM

San Jose, CA

Software Developer Co-op

Aug. 2026 - Present

- Contribute to **OpenRAG**, IBM's open-source Retrieval-Augmented Generation platform (**4,300+ GitHub stars**, built on **Langflow**, **Docling**, and **OpenSearch**) that powers agentic document search and AI-assisted chat over enterprise document collections.
- Own automated test coverage (**pytest**) for OpenRAG's **FastAPI** backend covering document ingestion and retrieval workflows, cutting manual regression-testing time by **14%** ahead of each release.
- Ship fixes and features across OpenRAG's **Python/FastAPI** backend and **TypeScript/Next.js** frontend, working directly with maintainers via code review on a **400+ fork** open-source repository.

JP Morgan Chase

Plano, TX

Software Engineer Intern

Jun. 2026 - Aug. 2026

- Built a full-stack **AI cost intelligence dashboard (React/TypeScript)** processing **3.1M+ rows** of LLM spend data through an offline **Node.js** ETL pipeline, supporting real-time multi-dimensional filtering without shipping raw data to the browser.
- Wrote a unit-tested (**Vitest**) analytics engine that computes spend anomaly detection, budget forecasting, and ranked cost-saving opportunities, used by finance stakeholders to review **FinOps** spend.
- Integrated **Databricks Genie** through a secure **Vite** server-side proxy and built client-side export to **Excel/PDF/PowerPoint**, cutting manual reporting steps for the cost-governance team.

PROJECTS

CourseMate | HTML, CSS, JavaScript, GraphQL API

Jan. 2026

- Built a **Chrome extension** that surfaces RateMyProfessors ratings directly inside UH's course registration flow, removing the need to tab-switch between the two sites.
- Developed an async web scraping system to fetch and cache ratings for **2,000+ UH professors**.
- Implemented local caching + rate limiting (**1 req/sec**) to cut external lookups by **30%**, reducing latency and staying within provider rate limits.

The Move – RSVP Component | React, TypeScript, Tailwind CSS, Node.js, JWT AuthMay 2025

- Designed and built an inline RSVP feature (Yes/No/Maybe) for a social event-sharing application.
- Developed real-time RSVP updates, load-tested to **10+ simultaneous users**.

E-Commerce Demand Forecasting System | Python, PyTorch, Flask, Docker, XGBoostMar. 2025

- Trained an **LSTM** model in **PyTorch** on **500K+ sales records** to forecast 30-day demand across **15 categories** of products selling **5,000 units/day**, achieving an **MAE of 2 units**.
- Built an **MLOps pipeline (Airflow + MLflow)** automating feature extraction from sales data and customer-review text (**BERT NLP**) for model retraining.
- Deployed an **XGBoost** classifier via **Flask API** and **Docker** to flag stockout risk for inventory restocking, evaluated with **89% precision** on held-out data.